

Customer Shopping Behavior Analysis

An end-to-end data analysis project exploring spending patterns, customer segments, and product preferences across **3,900 transactions** — using Python, PostgreSQL, and Power BI.



Dataset at a Glance

3,900

Total Purchases

Across all product categories

18

Data Columns

Demographics, purchases & behavior

37

Missing Values

In Review Rating only

25

Unique Items

Across 4 product categories

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color), and shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type).



Data Preparation in Python

01

Load & Explore

Imported dataset via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

02

Clean Missing Data

Imputed missing `Review Rating` values using median rating per product category.

03

Standardize & Engineer

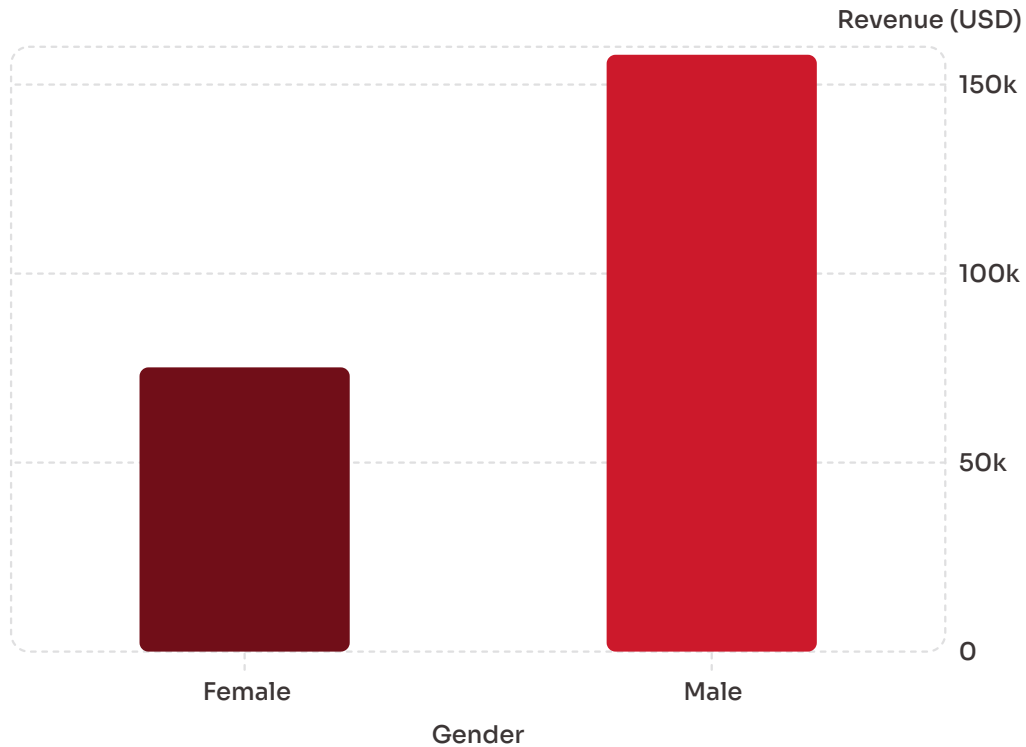
Renamed columns to `snake_case`; created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`.

04

Load to PostgreSQL

Connected Python script to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

Revenue by Gender



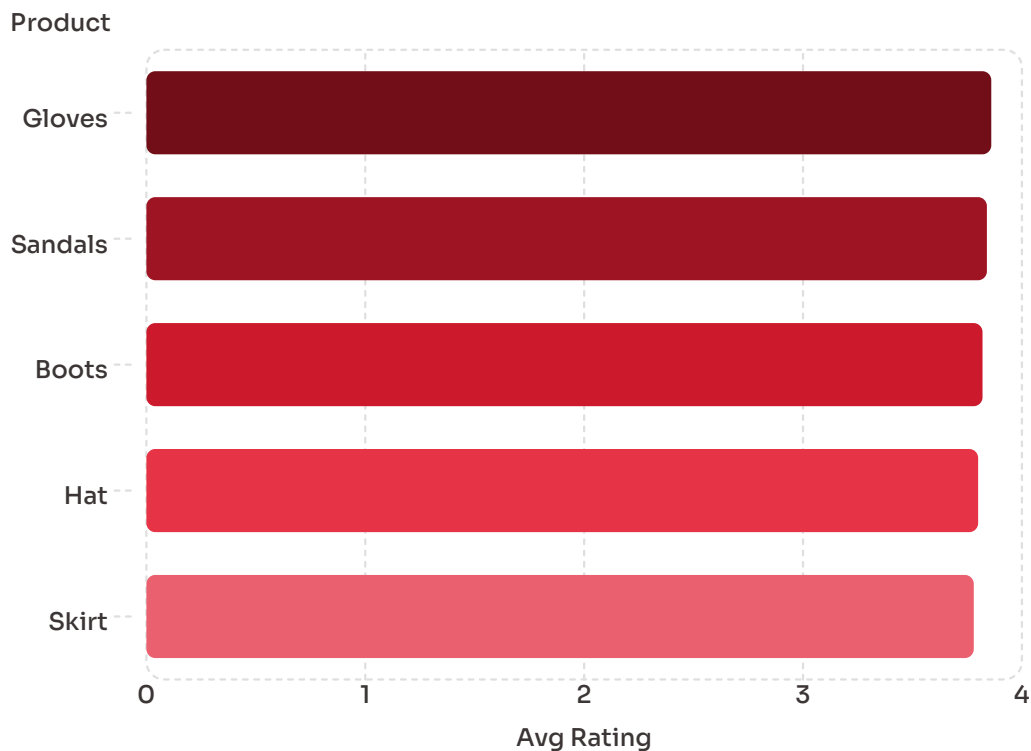
Male Customers Drive 2× More Revenue

Male customers generated **\$157,890** in total revenue compared to **\$75,191** from female customers — a significant gap worth investigating further.

- ❏ Male customers represent 2,652 of 3,900 transactions (68%), which partially explains the revenue disparity.

Product Ratings & Discount Dependency

Top 5 Products by Average Rating

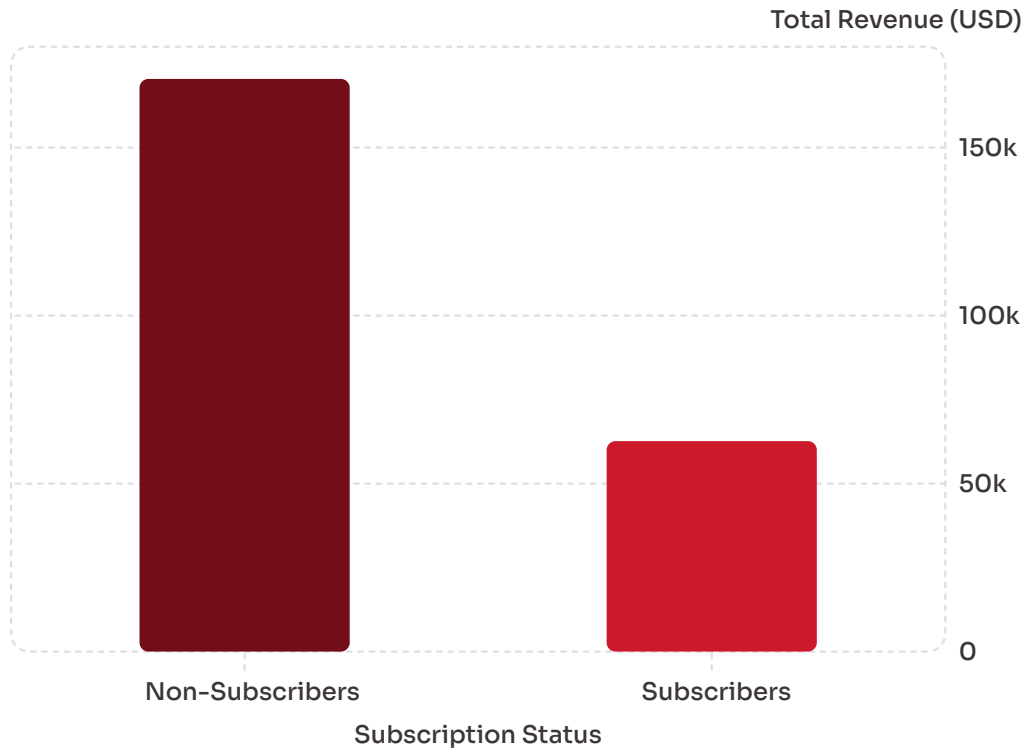


Most Discount-Dependent Products

Hats lead with **50%** of purchases made with a discount, followed by Sneakers (49.66%) and Coats (49.07%).

- Hat — 50.00%
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

Subscribers vs. Non-Subscribers



Key Findings

Non-subscribers generate **\$170,436** in total revenue vs. **\$62,645** for subscribers. However, average spend is nearly identical: **\$59.87** vs. **\$59.49**.

Of customers with more than 5 purchases, **958 are subscribers** and **2,518 are not** — suggesting repeat buyers are not necessarily converting to subscriptions.

- ❏ Subscribers are only 27% of the customer base but show strong repeat-purchase behavior.



Customer Segmentation

Loyal

3,116 customers — the dominant segment, representing the core revenue base.

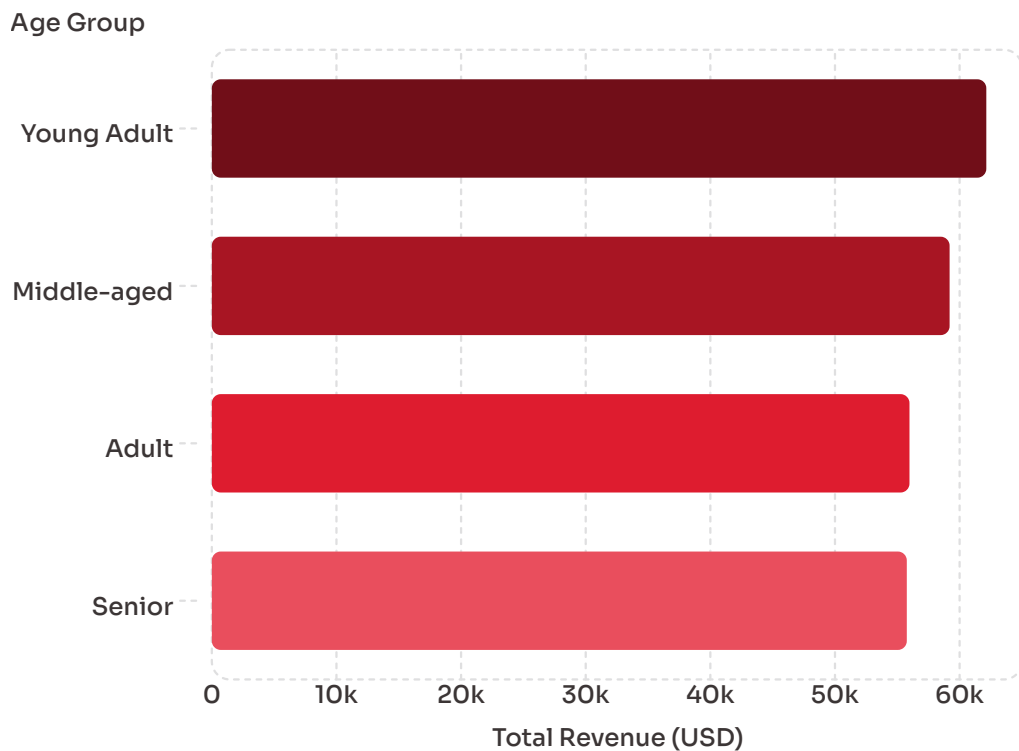
Returning

701 customers — engaged but not yet fully loyal; prime candidates for loyalty programs.

New

83 customers — smallest segment; onboarding and first-purchase incentives are critical.

Revenue by Age Group



Young Adults Lead Revenue

Young Adults contribute the highest total revenue at **\$62,143**, followed closely by Middle-aged customers at **\$59,197**. Revenue is relatively evenly distributed across all age groups, with only a **\$6,380** difference between the highest and lowest.

Express shipping users spend slightly more on average (**\$60.48**) than Standard shipping users (**\$58.46**), suggesting a willingness to pay for convenience.

Top Products by Category

Most purchased items within each of the four product categories:

Category	#1 Product	#2 Product	#3 Product
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

Clothing dominates overall with the highest purchase volumes, followed by Accessories. Footwear and Outerwear have lower but consistent order counts across their top products.

Power BI Dashboard & Recommendations

An interactive Power BI dashboard presents key metrics: **3.9K customers**, average purchase of **\$59.76**, and average rating of **3.75**. Filters include Subscription Status, Gender, Category, and Shipping Type. Revenue by Category shows Clothing leading at \$100K, followed by Accessories (\$75K), Footwear (\$30K), and Outerwear (\$15K).

→ **Boost Subscriptions**

Promote exclusive subscriber benefits to convert repeat buyers.

→ **Loyalty Programs**

Reward returning customers to move them into the Loyal segment.

→ **Review Discount Policy**

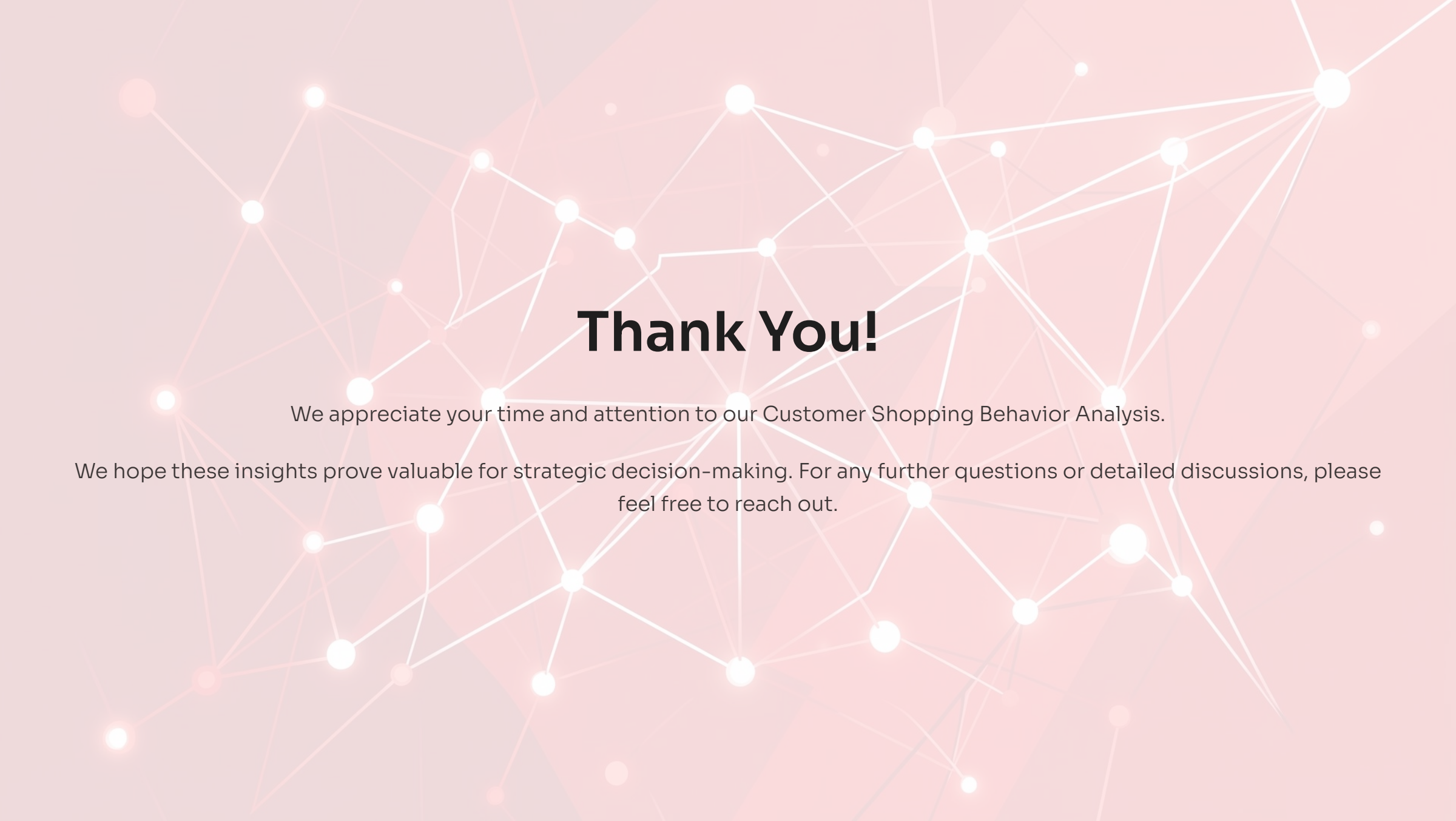
Balance sales boosts with margin control on discount-heavy products.

→ **Targeted Marketing**

Focus on high-revenue age groups and express-shipping users.

→ **Product Positioning**

Highlight top-rated and best-selling products in campaigns.



Thank You!

We appreciate your time and attention to our Customer Shopping Behavior Analysis.

We hope these insights prove valuable for strategic decision-making. For any further questions or detailed discussions, please feel free to reach out.